

Social Media Engagement Analysis

Tools: Python, Pandas, Matplotlib, Seaborn, Plotly

Dataset Size(When Provided): 5000 Rows

Dataset Overview : 5000 Rows data with columns :

user_id,age,gender,country,post_id,post_type,post_category,likes,comments,shares,watch_time_sec,impression_count,posted_at,follower_count,is_verified,device_type,sentiment,engagement_rate,hashtag_count,hashtags

Task 1 Data Import & Setup Details:

1. Checked for the data types whether they were correct or not
2. Converted date columns to datetime, posted_at is in the object(string) it converts to datetime64.
3. Used numpy arrays for basic array operations : np.array(), avg likes of an array is 10,105, std is 5703 etc

Task 2 Data Cleaning Details:

1. Detected the missing values using isnull(), isna()
2. Removed the missing values using dropna(), fillna(), median, mode, forwardfill() and backwardfill()
3. Find and remove duplicates using df.duplicated(), drop_duplicates()
4. Identified incorrect datatypes and fixed it.
5. Standardized the categories, here astype(str) convert to string, str.strip() used for removing the space, str.lower().str.title() to lower the Title case.
6. Corrected the unrealistic values using describe() and (df['likes']<0).sum().found 0 negative values in likes, comments, shares.
7. As part of feature cleaning first converted the Hashtag_Count column by splitting string and count tags convert into numeric.

Task 3: Data exploration using Pandas Details:

1. View dataset structure using head(), tail(), shape, and columns and got the result is Shape: (4991, 20), view the 20 columns, using head() show the first 5 posts, using tail showed the last 5 posts
2. Checked data types and info with info() and dtypes.
3. Generated summary statistics using describe().
4. Analyzed categorical distributions using value_counts(), unique(), and nunique(), on columns 'gender','country','post_type','device_type','post_category','sentiment','is_verified'.
5. Created a correlation matrix for numeric fields, using select_dtypes and corr(). Found weak correlations.
6. Used groupby() to summarize metrics here groupby() is used in the cases like Avg likes by post_Type, Avg impressions by country, Avg engagement by post category, this helps to identify top-performing content types and countries for business.

Task 4: Data Wrangling

1. Used **merge**, **concat**, or **join** if combining Data Frames. Splited the data frames into df1, df2 then used the concatenation function to join the data.
2. Create new fields such as engagement score, Hashtag Count
3. Performed group by Post _Type, Groupby by country, Groupby by sentiment for aggregating mean likes,comments,shares, engagement scores and post count , sorted engagement score descending .negative sentiment has highest engagement, video has more engagement, France leads as country wise.

Task 5: Statistical Analysis

1. Performed statistical analysis on columns Likes, Comments ,Shares ,watch time , Engagement Rate, Follower Count , here computed the mean, median, mode, std, variance, min/max,percentiles.

Task 6 : Data Visualization

1. Scatter : likes vs impressions→Showed the data visualization no linear correlation between likes and impression.
2. **Line:** *daily engagement trend → by grouping posted date and avg engagement score, no clear growth or seasonal trend.*
3. Bar: posts by category→ all 8 categories were balanced. Fitness is highest (~670),followed by tech,music,education etc... means a well balanced dataset.
4. Pie: gender distribution ->gender distribution is almost equal. Female vs male have only ~5% difference.
5. Count plot: post type ->perfectly balanced
6. Bar plot: avg likes by category -> all categories have avg likes
7. Violin: followers vs sentiment→all 3 sentiments shows similar distribution, means sentiment des not effects the follower count.
8. Heatmap: correlation matrix → all pair wise correlation is near to 0, indicating no strong linear relationships.
9. Plotly (Interactive-bubble)→no effective variation between follower count and engagement rate.

Final Insights: (Content Performance, User Trends and behavioural)

1. Highest Engagement rate is video
2. Best performing content category is music
3. Country having highest avg engagement rate is Brazil
4. 18-25 (youngsters) Age group have highest engagement rate.
5. Non verified 395K,verified 382K followers are not bother about verified accounts
6. Random hour (0-23) is the best time for views
7. Platform as device watch time is
8. Best sentiment performance is negative